

QUIZ 3 — Camera Calibration

Q41. Intrinsic camera parameters typically include:

Answer: B. Focal length, principal point, and skew

Q42. Extrinsic camera parameters describe:

Answer: B. The camera's rotation and translation relative to the world

Q43. In calibration, focal length is often expressed in units of:

Answer: B. Pixels

Q44. Radial lens distortion causes straight real-world lines in an image to appear:

Answer: B. Curved

Q45. "Barrel" distortion is characterized by straight lines that appear to:

Answer: B. Bow outward away from the centre

Q46. "Pincushion" distortion is characterized by straight lines that appear to:

Answer: B. Bow inward toward the centre

Q47. A commonly used calibration target for estimating camera parameters is a:

Answer: B. Checkerboard pattern

Q48. Reprojection error in camera calibration measures the difference between:

Answer: B. Observed image points and their reprojected estimated positions

Q49. The orthographic projection camera model essentially discards:

Answer: B. Perspective/depth effects

Q50. The weak-perspective camera model is a reasonable approximation when:

Answer: C. Depth variation across the object is small relative to viewing distance

Q51. The general camera projection matrix P has dimensions:

Answer: B. 3×4

Q52. Zhang's widely used calibration method requires:

Answer: B. Multiple images of a planar target at different orientations

Q53. Intrinsic and extrinsic parameters can be recovered from a projection matrix using:

Answer: A. RQ (or QR) decomposition

Q54. The principal point of a camera refers to:

Answer: B. The intersection of the optical axis with the image plane

Q55. The skew parameter in the intrinsic matrix accounts for:

Answer: C. Non-perpendicularity between the pixel axes

Q56. Tangential distortion in a lens typically arises due to:

Answer: B. Lens and sensor plane not being perfectly parallel (decentering)

Q57. Using more calibration images from varied viewpoints generally:

Answer: B. Reduces reprojection error and improves accuracy

Q58. The Direct Linear Transform (DLT) method is used to:

Answer: A. Estimate the projection matrix from 3D–2D point correspondences

Q59. The camera intrinsic matrix K primarily encodes:

Answer: B. Camera-internal parameters like focal length and principal point

Q60. Compared to the orthographic model, the full perspective camera model additionally accounts for:

Answer: B. Depth-dependent foreshortening (perspective effects)